# Linear Equations with Distribution and Fractions

Grade 8 / Pre-Algebra — Expressions and Equations

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**Skills practised:** Distributing first, clearing fractions with the LCD, variables on both sides, and equivalent rewrites.

**Worked pattern — follow this order every time.**

**Solve** $\frac{x}{2} + \frac{1}{3} = 2(x - 1)$.

**Step 1 — clear fractions.** The denominators are 2 and 3, so the LCD is 6. Multiply *every term on both sides* by 6: $3x + 2 = 12(x - 1)$.

**Step 2 — distribute.** Multiply through the brackets: $3x + 2 = 12x - 12$.

**Step 3 — collect.** Variable terms to one side, numbers to the other: $14 = 9x$.

**Step 4 — isolate and check.** $x = \frac{14}{9}$. Substitute back into the *original* equation to check.

Show every step. Leave answers as exact fractions where they are not whole numbers.

**1.** Solve for $x$: $3(x + 4) = 21$

Answer: ____________________

**2.** Solve for $x$: $-2(x - 5) = 14$

Answer: ____________________

**3.** Solve for $x$: $\dfrac{x}{3} + \dfrac{x}{6} = 5$

(Start by multiplying every term by the LCD.)

Answer: ____________________

**4.** Solve for $x$: $4 - 2(x + 3) = 10$

(The subtraction sign in front of the brackets belongs to the 2.)

Answer: ____________________

**5.** The expression $\dfrac{x}{4} + \dfrac{2}{3}$ appears on one side of an equation. Multiply it by the LCD, 12, and write the equivalent expression with no fractions left in it.

Answer: ____________________

**6.** Solve for $x$: $3(x-2) = 2(x+4)$

Answer: ________________

**7.** Solve for $x$: $\dfrac{3x}{4} - 2 = \dfrac{x}{2}$

Answer: ________________

**8.** Solve for $x$: $\dfrac{1}{2}(6x-8) = 11$

Answer: ________________

**9.** Expand and simplify $7 - 3(2x - 5)$, writing your answer in the form $a + bx$ or $bx + a$.

Answer: ____________________

**10.** Solve for $x$: $\frac{2}{3}(x + 6) = \frac{1}{2}(x + 2) + 5$

Answer: ____________________

**11.** Solve for $x$: $\frac{x + 3}{2} - \frac{x - 5}{4} = 5$

(A fraction bar groups the whole numerator — keep the brackets when you multiply.)

Answer: ____________________

**12.** **Challenge.** Solve for $x$, then check your solution in the original equation: $\frac{2}{5}(10x - 15) = 3(x-1) + 4$

Answer: __________________